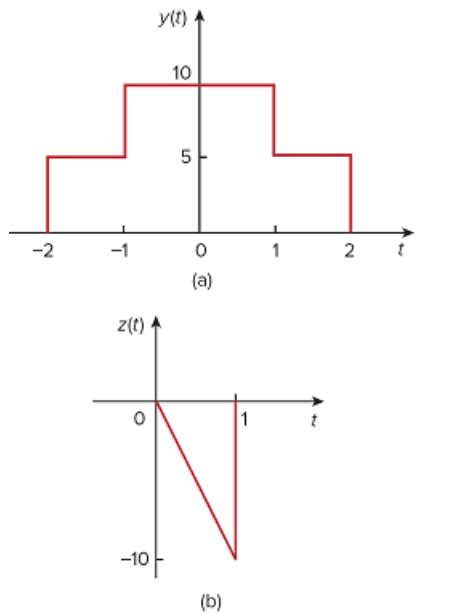


Problem

Obtain the Fourier transforms of the signals shown in Fig. 18.33.

Figure 18.33



Step-by-step solution

Step 1 of 6

(a)

(Refer to Figure 18.33 (a) in the textbook) the signal $f(t)$ is represented as,

$$f(t) = \begin{cases} 12, & 0 < t < 1 \\ (24 - 12t), & 1 < t < 2 \end{cases}$$

Step 2 of 6

Take the Fourier transform of signal $f(t)$.

$$F(\omega) = 12 \int_0^1 e^{-j\omega t} dt + 24 \int_1^2 e^{-j\omega t} dt - 12 \int_1^2 t e^{-j\omega t} dt$$

The integration of $x e^{ax}$ is,

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Then, the signal $F(\omega)$ is,

$$\begin{aligned} F(\omega) &= 12 \int_0^1 e^{-j\omega t} dt + 24 \int_1^2 e^{-j\omega t} dt - 12 \int_1^2 t e^{-j\omega t} dt \\ &= 12 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 + 24 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_1^2 - 12 \left[\frac{e^{-j\omega t}}{(-j\omega)^2} (-j\omega t - 1) \right]_1^2 \\ &= \frac{-12}{j\omega} [e^{-j\omega} - 1] + \frac{24}{-j\omega} [e^{-2j\omega} - e^{-j\omega}] - 12 \left[\frac{e^{-2j\omega}}{\omega^2} (2j\omega + 1) - \frac{e^{-j\omega}}{\omega^2} (j\omega + 1) \right] \\ &= \frac{12j\omega e^{-j\omega} - 12j\omega + 24j\omega e^{-2j\omega} - 24j\omega e^{-j\omega} - 24j\omega e^{-2j\omega} - 12e^{-2j\omega} + 12j\omega e^{-j\omega} + 12e^{-j\omega}}{\omega^2} \end{aligned}$$

Step 3 of 6

Rewrite the above equation.

$$F(\omega) = \frac{-12j\omega - 12e^{-2j\omega} + 12e^{-j\omega}}{\omega^2}$$

Hence, the Fourier transform of the signal $f(t)$ is $\boxed{\frac{-12j\omega - 12e^{-2j\omega} + 12e^{-j\omega}}{\omega^2}}$.

Step 4 of 6

(b)

(Refer to Figure 18.33 (b) in the textbook) the signal $g(t)$ is represented as,

$$g(t) = \begin{cases} 6, & -2 < t < -1 \\ 3, & -1 < t < 1 \\ 6, & 1 < t < 2 \end{cases}$$

Step 5 of 6

Take the Fourier transform of signal $g(t)$.

$$\begin{aligned} G(\omega) &= 6 \int_{-2}^{-1} e^{-j\omega t} dt + 3 \int_{-1}^1 e^{-j\omega t} dt + 6 \int_1^2 e^{-j\omega t} dt \\ &= 6 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-2}^{-1} + 3 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^1 + 6 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_1^2 \\ &= \frac{-12}{\omega} \left[\frac{e^{j\omega} - e^{-j\omega}}{2j} - \frac{e^{2j\omega} - e^{-2j\omega}}{2j} \right] + 6 \left[\frac{e^{j\omega} - e^{-j\omega}}{2j} \right] \\ &= \frac{-12 \sin \omega}{\omega} + \frac{12 \sin 2\omega}{\omega} + \frac{6 \sin \omega}{\omega} \end{aligned}$$

Step 6 of 6

Rewrite the above equation.

$$G(\omega) = \frac{6 \sin 2\omega}{\omega} - \frac{3 \sin \omega}{\omega}$$

Hence, the Fourier transform of signal $g(t)$ is $\boxed{\frac{12 \sin 2\omega}{\omega} - \frac{6 \sin \omega}{\omega}}$.